V05. Safety supervision for a 500 m^3 swimming pool

Stateflow / Simulink R2026a | October 23, 2026 | Seven simulated cases, 168 hours each, 900-second step

Evidence boundary: this is a fault-injection experiment using synthetic operating inputs and virtual sensors. It is NOT independent hardware verification, SIL/PL certification, or evidence of regulatory compliance.

1. System architecture

The new v05 model is a separately saved version of the closed-loop v04 model. Neither v04 nor the original Stateflow v85 was changed. Two simulated water-temperature sensors, flow confirmation, emergency-stop and manual-reset commands enter a supervisory gate **before** the pool heat-demand and allocation controller. A detected fault disables only pool heating. Independent data-centre waste-heat rejection continues to satisfy the energy balance.

2. Dynamic and safety mathematics

- Pool thermal capacitance: **C = 997 × 500 × 4182 = 2,084,727,000 J/K**.
- Water energy: T(k+1) = T(k) + 900(Q_pool - Q_loss)/C.
- Sensor models: T1 = T_model + bias1; T2 = T_model + bias2.
- Disagreement: abs(T1 - T2) > 0.35 K.
- Plausibility in this simulated model: abs(bias1) > 0.45 K or abs(bias2) > 0.45 K.
- Overtemperature: max(T1,T2) >= 27.70°C.
- Other faults: no flow, emergency stop, or nonfinite sensor values.
- Any active fault causes a **latched shutdown**, with Q_pool = 0.
- Manual reset is accepted only after all monitored faults have cleared and modeled water T <= 27.65°C.
- Data-centre first law: Q_DC = Q_HX_direct + Q_HP_source + Q_REJECT.
- Heat-pump first law: Q_HP_out = Q_HP_source + P_compressor.

These thresholds are assumed design parameters, not measured sensor specifications or normatively certified setpoints.

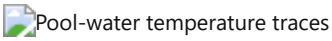
3. Executed scenarios

Scenario	Locked [h]	Pool heat [MWh]	Successful resets
No fault	0	10.393	0
Sensor disagreement	8.00	10.197	1
Slowly drifting sensor	13.75	10.343	1
Flow loss	14.00	10.352	1
Emergency button	18.00	10.311	1
Reset attempted during fault	18.00	10.255	1, after clearance
Equal bias on both sensors	15.00	10.359	1

All seven MATLAB/Simulink runs completed. Independent Python validation passed **32,928 assertion groups** covering the thermal conservation, pool shutdown interlock, source and heat pump balances, latch sequencing and explicit reset. Maximum simulated temperature was 27.58°C. The healthy run reproduces v04 within tolerance 10^-7.

4. Rendered simulation evidence


Fault latch timeline


Pool-water temperature traces

5. Limitations and next engineering stage

Both simulated sensor channels share the true state of the mathematical model; detecting a common-mode sensor error in this experiment does **not** demonstrate common-mode fault detection in a physical plant. Required next: independent field sensors, response-time modeling, stuck-open valve and cooling-failure injection, heat exchanger hydraulics, manufacturer COP maps, measured data and current consolidated compliance review.

References: EU Delegated Regulation 2024/1364, STN EN 15288-1+A1:2024 (applicability to be confirmed), and the historical Slovak 308/2012 text (which specifies a 28°C maximum for a swimming pool in §8(7); current legislation requires separate confirmation).